

Foreword

You did it! You've made it to the end of Linear Algebra! Because this is a final review, there is no Quick Review. I wanted to say a quick thank you to everyone whose been a part of the AEWs this semester. It has been an absolute blast to work with you all and I wish you the best as you go into this final. Good luck and I hope these AEWs have been helpful!

Problems

1. A matrix transformation $T : \mathbb{M}_{2 \times 2} \rightarrow \mathbb{P}^3$ is defined as shown below.

$$\begin{bmatrix} a & b \\ b & c \end{bmatrix} \mapsto ax^3 + bx^2 + cx + (a + b + c)$$
$$a, b, c \in \mathbb{R}$$

- (a) **(PROOF)**. Let S represent the space spanned by the domain. Prove that S is a subspace of $\mathbb{M}_{2 \times 2}$.

- (b) **(PROOF)**. Prove that this is an isomorphism.

- (c) Use the standard inner product on $\mathbb{P}^3$ to determine $\langle \mathbf{U}, \mathbf{V} \rangle$ where

$$\mathbf{U} = \begin{bmatrix} 2 & 3 \\ 3 & 5 \end{bmatrix}, \mathbf{V} = \begin{bmatrix} 4 & 1 \\ 1 & 9 \end{bmatrix}$$

2. Below is a matrix $\mathbf{A}$.

$$\mathbf{A} = \begin{bmatrix} 6 & -a \\ 2 & a \end{bmatrix}$$

(a) Find the smallest value of a that produces a *repeated* eigenvalue. This will be the value you use for a in the later parts. (Hint: Use the characteristic equation to find repeated roots!)

(b) Determine the single eigenvalue of $\mathbf{A}$. Also, show that this is a defective matrix.

(c) Find an orthonormal basis for the column space of $\mathbf{A}$.

(d) Determine the row space and null space of the matrix $\mathbf{A}$.

3. Below is a symmetrical matrix $\mathbf{A}$.

$$\mathbf{A} = \begin{bmatrix} -1 & 2 & 0 \\ 2 & 0 & 2 \\ 0 & 2 & 1 \end{bmatrix}$$

(a) Find matrices $\mathbf{D}$ and $\mathbf{P}$ such that $\mathbf{D} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$.

(b) Using your result from part (a), determine $\mathbf{P}^{-1}\mathbf{A}^5$.

4. Let $T : \mathbb{F}(-\infty, \infty) \rightarrow \mathbb{F}(-\infty, \infty)$ be defined as

$$T(f(x)) = f(x + 2)$$

- (a) If we were to apply this transformation to a function $g(x) = \frac{1}{x}$, we would find that this transformation appears to be non-linear. Why does this case not matter?

- (b) Show that T is a linear transformation.

5. Let there be two bases in $\mathbb{R}^2$: $A = \{(3, a), (5, 2)\}$, $B = \{(6, 3), (b, 2)\}$, where $a, b \in \mathbb{R}$.

- (a) Determine the value b for which the transformation matrix $P_{A \rightarrow B}$ does not exist.

- (b) Determine the value a for which the transformation matrix $P_{B \rightarrow A}$ does not exist.

- (c) If $P_{A \rightarrow B}$ exists, does that imply that $P_{B \rightarrow A}$ also exists?

6. **(BONUS)**. If $\mathbf{A}$ is a square matrix and $p(\lambda) = \det(\lambda\mathbf{I} - \mathbf{A})$ is the characteristic polynomial of $\mathbf{A}$, then the coefficient of λ^{n-1} in $p(\lambda)$ is the negative of the trace of $\mathbf{A}$. **(Taken from Chapter 5 Supplementary Exercise #5)**.